

No. of Functional Features Included in the Solution

Date	1 July 2026
Team ID	—
Project Name	Rising Waters – Flood Prediction System
Maximum Marks	5 Marks

This document lists all the functional features included in the Rising Waters solution.

Functional Features Overview

S. No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Dataset Generation	Simulates a realistic flood dataset with missing values and outliers	notebooks/0_generate_dataset.py	Done	0.5
2	Exploratory Data Analysis	Distribution, boxplot, heatmap and pairplot visualizations	notebooks/1_data_visualization.py	Done	0.5
3	Data Preprocessing	Median imputation, IQR outlier capping, train/test split, scaling	notebooks/2_preprocessing_and_training.py	Done	1
4	Multi-Model Training	Trains Decision Tree, Random Forest, KNN and XGBoost classifiers	notebooks/2_preprocessing_and_training.py	Done	1
5	Best Model Selection & Persistence	Selects highest-accuracy model, saves model/scaler/metadata as artifacts	notebooks/2_preprocessing_and_training.py	Done	0.5
6	Home Page	Displays project overview, active model and accuracy	templates/home.html	Done	0.5
7	Prediction Form	Collects 10 weather/terrain inputs with labels, units and defaults	templates/predict.html	Done	0.5
8	Flood Risk Prediction	Scales input, runs model, returns Flood/No-Flood + probability	app.py	Done	1

Feature Summary

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8

Metric	Count / Value
Core / Must-Have Features	6 (dataset, preprocessing, training, selection, form, prediction)
Additional / Nice-to-Have Features	2 (EDA visualizations, home page model display)
Features Tested & Verified	8

Feature Category Breakdown

S. No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Home page, prediction form, result pages
2	Backend / Logic	3	Flask routing, scaling/prediction logic, error handling
3	Database / Storage	1	CSV dataset + pickled model artifacts
4	API / Integration	0	Not applicable (self-contained app)
5	Security / Authentication	0	Not applicable (no login/PII in this scope)